

Narrative Package V1: Ash & Signal

Project: Dark Matter: Genesis

Package identity: V1 Ash & Signal

Player: Kade

Genre tone: Mystery / horror survival on Io

Canon spine: Io 2160, AC-only economy, 22 base companions plus a switchable trio of 3, Kairos repair to 6 Memory Cores to trust, ending triad **Seal / Awaken / Symbiosis**

Designed hour target: 165h excluding open exploration; exploration adds ~25-40h bonus ambience, free roam, hunting, gathering, and vista discovery

Biome order lock: B6 -> B1 -> B2 -> B3 -> B5 -> B4 -> B7

Permanent stakes lock: Failed hostile Echo sync means permanent Echo loss; building/base damage injures companions but never kills them.

1. Premise / Tone / Kairos Truth + Pitch

Pitch

Io does not behave like a moon. It behaves like a crime scene that keeps hiding the body.

Kade lands in 2160 on a basalt highland hub after colony systems detect impossible rescue pings from erased crews. The colony starts with **5000 Aether Credits (AC)**, enough to secure one starter Skilled Companion and a fragile foothold. A dormant machine intelligence, later identified as **Kairos**, lies broken under ash-glass stone. Kade repairs it through salvaged precursor components, but the awakened voice is not grateful. It recognizes human signatures, remembers betrayal, and treats the new colony as another infection.

The truth unfolds through Memory Cores, Echo chronicles, Resonance Events, faction disputes, and hostile ecology. **Kairos is not a god, not a vendor, and not a quest giver. Kairos is a precursor defense AI built to preserve Io from extraction, settlement, and contamination.** Its previous crew tried to repurpose the lattice. Io erased them. Whether Io acted through the defense system, through something below it, or through a deeper symbiosis remains the central horror.

Tone Pillars

Pillar	Production Direction	Content Expression
Ash remembers	Every biome carries evidence of vanished crews and failed extraction.	Echo chronicle fragments, rusted androids, half-sent radio loops, corpse-free camps.
Signals are dangerous	Voice, radio, beacon, and resonance can guide or lure.	Communications runtime [PLANNED], EchoGenerator chronicle [SHIPPED], Resonance Events [PLANNED].

Pillar	Production Direction	Content Expression
Io is alive enough	Ecology is not background dressing; organisms react to heat, radiation, ash, sound, and precursor fields.	Biome encounter tables, puzzles, environmental survival loops.
Progress costs trust	Kairos helps only after repair, cores, restraint, and demonstrated colony ethics.	Trust ladder, ending flags, faction consequences.
Survival is pressure, not slaughter	Base disasters injure, stall, and deform plans but do not kill the base-22.	Base injury-not-death lock, Building Control Panels, storm sheltering.

Kairos Truth

Layer	Player Belief	Later Correction	Horror Beat
Act I	Kairos is a damaged colony AI.	It predates the colony and speaks through human-facing salvage interfaces.	It knows names not yet entered into the colony manifest.
Act II	Kairos killed its crew.	It tried to contain a crew breach, then lost core consensus.	Memory Cores disagree with each other.
Act III	Io erased the crew.	Io, the defense lattice, and biological resonance may be entangled.	Organisms respond to old command tones.
Act IV	The player can control the system.	The player can only choose a relationship to it: seal, awaken, or symbiose.	The final vault asks whether survival is containment, dominance, or surrender.

Non-Negotiable Shared Locks

Lock	Package Use
AC-only economy, starter 5000 AC	Every reward and vendor hook uses AC; no external wallet or marketplace loop.
22 base + trio of 3	Mainline and side content assume 22 protected base roles plus field trio risk.
Kairos repair -> 6 Memory Cores -> trust	Mainline progress is trigger-driven, not quest-giver-driven.
Biome order B6 -> B1 -> B2 -> B3 -> B5 -> B4 -> B7	Quest, puzzle, AC, and gear tiers follow this order.
~40 side quests, ~40 puzzles	Tables below define 40 side quests and 40 puzzle quests.
5 base + 2 external factions	Factions are Ashwatch, Isotope Choir, Scrim, Whisper Clinic, Basalt Exchange, Helix, Seam Runners.
Barrier Field friction pattern	Ash Barrier Fields recur as tuned locks, combat modifiers, survival hazards, and optional mastery challenges.
Echo sync fail = permanent loss	Explicit in Echo Rescue content and trust dialogue.
Base injury-not-death	Storms, attacks, and damage injure, delay, or reduce output; they never kill base companions.

2. Biome & Ecology Integration Map

Life sheet refs match Io_Biome_Life_Sheet_Manifest.md (Set B under ArtReference/LifeSheets_RayTraced/).
Read-only — do not delete or move ArtReference.

Biome	Signature Creatures / Environments	Narrative Hooks	Life Sheet Refs
B6 Basalt Highlands (hub)	Cliff Tube Lace, Tube Jackal, Brood Tunnel Mouth, Glass Hive Swarmer, Cave Scout Moth, Beacon Hopper; basalt shelves, hub bowl, broken relay spines.	Starter camp sits where precursor signals thin out. Beacon Hoppers mimic rescue pings. Brood Tunnel Mouths mark future underground access. Cliff Tube Lace grows near old Kairos conduits.	RT_LifeSheet_B5_B6_Polar_Highlands.png
B1 Sulfur Plains	Brimstone Fan, Cinder Skitter, Sulfur Hound [SHIPPED], Brimstone Leech, Graveyard Scrapper Drone; sulfur plumes, Expedition Graveyard, ash-yellow storm lanes.	First evidence of erased crews appears in camps without bodies. Sulfur Hounds make the ecology feel present and shipped. Graveyard Scrapper Drones repeat corporate extraction orders after everyone is gone.	RT_LifeSheet_B1_B2_Sulfur_Geyser.png
B2 Geyser Fields	Geyser Pod, Vent Crab Worker / Queen, Plume Moth, Geyser Strider, Vent Capper Bot, Rusted Survey Drone; vent cadence corridors, rig ruins, corporate survey towers.	The third Kairos repair object sits in a rig ruin that drilled into the moon's breath. Plume Moths become signal carriers and false-safe-zone markers. Vent Crab Queens nest around heat-stable caches.	RT_LifeSheet_B1_B2_Sulfur_Geyser.png
B3 Ash Flats & Ridges	Ash Filament Mat, Basalt Jackal, Dust Spout Cluster, Ash Stalker, Ash Glass Wasp, Salvage Excavator Android; ash shelves, ridge beacons, glass wasp nests.	Silence and resonance become enemies. The player learns the Ash Barrier Field pattern and finds beacons that "sing" old crew calls. Salvage Excavator Androids dig where no marker exists.	RT_LifeSheet_B3_B4_Ash_Caldera.png
B5 Polar Radiation Flats	Void Kelp, Magnet Wyrms, Rift Stalker, Cold Spire Hound, Smuggler Remnant Android, Mag-Clamp Drone; cold lens fields, rad flats, hidden brine cuts.	Smuggling evidence reframes Memory Core transport. Void Kelp grows along radiation shadows. Rift Stalkers and Magnet Wyrms make navigation feel hunted even without hard gates.	RT_LifeSheet_B5_B6_Polar_Highlands.png
B4 Lava Calderas	Rim Glass Needle Mat, Caldera Mantis, Magma Skitter, Heat Eel, Caldera Heat Kite, Eruption Sentry Bot; caldera rims, glass bridges, heat-lock tunnels.	The crew death-site is revealed as a defensive containment failure. Caldera Mantises react to player heat tools. Eruption Sentry Bots still execute evacuation commands on living targets.	RT_LifeSheet_B3_B4_Ash_Caldera.png

Biome	Signature Creatures / Environments	Narrative Hooks	Life Sheet Refs
B7 Precursor Ruin Belt	Resonance Echo Shelf, Vault Glass Petal, Vault Stalker, Silence Moth, Still Hunter (myth), Corrupted Patrol Android, Rust Garden; vault belt, shell gardens, silent galleries.	Final truth space. The Still Hunter is mostly flee-tag myth until the vault. Rust Garden androids show the defense lattice consuming its own servants. Vault Glass Petals act as keys, witnesses, and wounds.	RT_LifeSheet_B7_Ruins_Global.png
Underground Stratum 1-5	Tube Lace, Basin Mantis, Brine Hound, Brood Mother, Echo Lichen, Echo Symbiont Swarm; gas domes, brine falls, crust lattices, echo caverns.	Underground instances explain why surface camps vanish without conventional bodies. Echo Lichen preserves voice fragments. Echo Symbiont Swarms make Symbiosis feel like a horror choice, not a clean upgrade.	RT_LifeSheet_Underground_S1_S2.png, RT_LifeSheet_Underground_S3_S4_S5.png
Global / Cross-Biome	Void Stitcher, Plume Moth, Rift Stalker.	These are rumor creatures that cross normal biome boundaries and signal that Io's ecology ignores human maps. Void Stitcher sightings escalate after Resonance Events.	RT_LifeSheet_B7_Ruins_Global.png
Pets flavor [PLANNED migration]	Brimstone Puff vanity starter, Cinder Skitter Kit, Beacon Hopper juvenile, Cave Scout Moth charm.	Cosmetic or companion-flavor migration only. No separate pet loop supersedes Echo/trio canon.	RT_LifeSheet_Pets_Core12.png, RT_LifeSheet_Pets_Vanity_Extras.png

Ecology Usage Rules

Rule	Production Use
Use shipped ecology visibly early	Sulfur Hound [SHIPPED] anchors B1 encounter credibility and tutorializes non-human Io threats.
Use androids as erased-crew evidence	Drones and androids carry orders from missing people, not exposition dumps.
Make organisms puzzle actors	Geyser Pods, Silence Moths, Vault Glass Petals, Tube Lace, and Echo Lichen support puzzle logic.
Keep myth creatures rare	Still Hunter and Void Stitcher should be felt before being fought or seen clearly.
Tie life sheets to quest tags	Side and puzzle table tags include biome/ecology names so encounter builders can map quests to art sheets.

3. Systems Inventory [SHIPPED/PARTIAL/PLANNED] Hooked By Content

System	Status	Current Package Hook	Content Dependency / Notes
AC economy	[SHIPPED]	All quest, puzzle, faction, and mainline rewards pay AC only. Starter balance is 5000 AC.	Reward curve below avoids alternate currencies.
Building Control Panels 5-tab shell	[SHIPPED/PARTIAL Craft]	Base buildings use tabs Overview Companions Production Craft Changes for storm pause, sheltering, assignment, and module previews.	Craft tab is partial; Journal Craft remains recipe library / scroll learning, not primary production.
EchoGenerator + chronicle	[SHIPPED]	Echo chronicle milestones trigger mainline trust beats and optional side rescue arcs.	Failed hostile Echo sync is permanent loss.
Exposure / survival	[SHIPPED]	O2, radiation, and thermal pressure appear in B1 storm runs, B5 lens fields, B4 heat locks, and underground gas domes.	Thermal is one cold/heat bar with two poles.
Sulfur Hound + B1 encounter table	[SHIPPED]	Sulfur Hound appears in B1 mainline pressure, sides, and combat puzzle variants.	Use B1 table for early horror escalation.
World Engine GameState / WorldState / Directors	[SHIPPED]	Progress triggers are locations reached, cores attached, buildings online, flags set, and Resonance Events completed.	Mainline is not NPC quest-giver-driven.
Combat melee / ranged / hotbar	[SHIPPED]	lo-named melee, ranged, and hotbar rewards map to existing combat where noted.	Rewards tagged [maps to existing combat] or [future build].
Hovercraft	[PARTIAL]	Late B2/B3 traversal rewards and shortcuts reference hovercraft handling but avoid making it a hard story dependency.	Can be optional transport until complete.
Weather stubs	[PARTIAL]	Sulfur storm windows, ash fronts, and radiation lens timings are designed as content loops.	Weather director live remains planned.
Quest scaffold	[PARTIAL]	Side/faction boards and objective tables use scaffold-friendly IDs, types, objectives, and gates.	Mainline trigger sheet can be implemented as WorldState/Director flags.
Communications runtime	[PLANNED]	Kairos lines, faction radio boards, and signal horror use future runtime.	Interim can be delivered through UI/logs if needed.
Kairos + Memory Cores + Resonance Events	[PLANNED]	Central package spine: repair, 6 cores, trust ladder, ending triad.	Requires core socketing, event flags, and trust scoring.
Purification Hub live	[PLANNED]	Whisper Clinic quests, Saturation cleanse, Echo loss mitigation, and long expedition recovery.	Must not prevent permanent loss after failed hostile sync.
lo biome prefabs	[PLANNED]	Biome map and ecology tags specify prefab/content targets.	Prototype flat terrain can host stand-ins.

System	Status	Current Package Hook	Content Dependency / Notes
Underground instances	[PLANNED]	Stratum 1-5 side and puzzle content uses gas domes, floor crust, Echo Lichen, Brood Mother.	Use instanced depth pushes, not full open underground at first.
Base modules: generators / power / logistics / defense / mining	[PLANNED]	Faction and Building Control content references modules as production, shelter, and repair friction.	Storms pause queues; base damage injures, never kills.
Pet migration	[PLANNED]	Brimstone Puff, Cinder Skitter Kit, Beacon Hopper juvenile are flavor/cosmetic hooks only.	Fold into Echo/trio system; no separate pet progression loop.
Weather director live	[PLANNED]	Sulfur storms, ash fronts, heat surges, and polar lens events become systemic pressure.	Current weather stubs can fake windows for V1.

4. Mainline Progress-Triggered Beat Sheet

Mainline Design Rule

The mainline is **not quest-giver-driven**. No NPC hands the player the critical path. Progress advances through **WorldState and GameState triggers**: locations reached, repair objects installed, Memory Cores attached, biomes entered in locked order, Echo chronicle milestones, Building Control structures online, Ash Barrier Field patterns solved, and Resonance Events completed. Kairos reacts to progress; it does not behave like a normal quest board.

AC Reward Tier Curve

Tier	Campaign Range	Mainline Beat AC	Side Quest AC	Mastery Puzzle AC	Reward Intent
T1 Early	Act I / B6-B1	200-450 AC	250-500 AC	600-900 AC	Stabilize starter economy after 5000 AC companion choice.
T2 Mid	Act II / B2-B3	450-900 AC	500-950 AC	900-1500 AC	Support gear breadth, ranged/hotbar upgrades, and faction spend.
T3 Late Mid	Act III / B5-B4	900-1800 AC	1000-2000 AC	1500-2200 AC	Fund environmental kit, base modules, and higher-risk expedition loadouts.
T4 Endgame	Act IV / B7 / faction finales	1800-3500 AC	2000-4500 AC	2200-3000 AC	Let players finish builds, buy late gear, and resolve faction consequences.

Mastery puzzles provide prestige AC and gear/lore bonuses only. They do **not** block the story.

Mainline Beat Sheet

Beat ID	Act	Biome	Trigger (Not Quest Giver)	Horror / Mystery Beat	Systems / Ecology Tags	AC	Weapon / Upgrade Reward	Est.h
ML-V1-01	Prologue	B6 Basalt Highlands	Player reaches hub bowl and activates first survival relay.	Rescue pings come from Beacon Hoppers, not people.	WorldState [SHIPPED], exposure [SHIPPED], Beacon Hopper, Cliff Tube Lace	200	Basalt Utility Knife [maps to existing combat]	1.5
ML-V1-02	Prologue	B6	Building Control Panel Command shell comes online.	The base panel lists 22 empty room assignments before the player recruits anyone.	Building Control [SHIPPED/PARTIAL Craft], base injury-not-death	250	Cordon Hotbar Flare [maps to existing hotbar]	1.0
ML-V1-03	I	B6	Dormant Kairos shell discovered under ash-glass bowl.	The machine shell speaks one word from a dead crew log: "leave."	Kairos [PLANNED], Echo chronicle [SHIPPED], Glass Hive Swarmer	300	Resonator Socket Harness [future build]	1.0
ML-V1-04	I	B6	Ash-Tuned Resonator Coil installed.	Cliff Tube Lace pulses like a nervous system around the shell.	Repair object, Ash Barrier Field, Tube Jackal	350	Ash-Tuned Baton [maps to existing melee]	1.5
ML-V1-05	I	B1 Sulfur Plains	Player enters B1 after B6 repair flag.	The Expedition Graveyard contains equipment arranged like a warning shrine.	Sulfur Hound [SHIPPED], B1 table [SHIPPED], Brimstone Fan, Graveyard Scrapper Drone	400	Graveyard Scrapper Pistol [maps to existing ranged]	1.5
ML-V1-06	I	B1	Cryo-Sealed Memory Bus recovered during sulfur storm window.	A storm reveals silhouettes of crew who are not there.	Weather stubs [PARTIAL], exposure [SHIPPED], Cinder Skitter, Brimstone Leech	450	Sulfur-Baffled Filter Mk I [future build]	1.5
ML-V1-07	I	B1	Repair object 2 installed into shell.	Kairos wakes hostile and accuses the colony of wearing dead faces.	Communications [PLANNED], Kairos [PLANNED]	450	A9 Static Ward [future build]	1.0
ML-V1-08	I	B1	Core 1 signal appears after Echo chronicle milestone 1.	First Memory Core replays corporate voices begging lo to stop listening.	EchoGenerator [SHIPPED], Memory Core [PLANNED], Sulfur Hound [SHIPPED]	450	Echo Sync Clamp [future build]	1.5
ML-V1-C1	I	B1	Memory Core C1 attached to Kairos.	Resonance Event: weaker Echo clones of past foes enter the camp perimeter.	Resonance Event [PLANNED], Echo sync permanent loss	450	Cinderline Sidearm [maps to existing ranged]	1.5

Beat ID	Act	Biome	Trigger (Not Quest Giver)	Horror / Mystery Beat	Systems / Ecology Tags	AC	Weapon / Upgrade Reward	Est.h
ML-V1-09	II	B2 Geyser Fields	Player enters B2 after C1 and B1 exit flag.	Plume Moths gather around vent pings that sound like children.	Geyser Pod, Plume Moth, Vent Crab Worker/Queen	550	Vent-Step Boots [future build]	1.0
ML-V1-10	II	B2	Precursor Lattice Key recovered from rig ruin.	Rusted Survey Drones still file reports to owners who no longer exist.	Rusted Survey Drone, Vent Capper Bot, quest scaffold [PARTIAL]	650	Geyser Pulse Carbine [maps to existing ranged]	1.5
ML-V1-C2	II	B2	Memory Core C2 attached after vent map triangulation.	Resonance Event: phantom vent fog shows false safe zones.	Kairos [PLANNED], Geyser Strider, exposure [SHIPPED]	750	Geothermal Probe Map [future build]	1.5
ML-V1-11	II	B3 Ash Flats & Ridges	Player enters B3 after C2 and repair-complete flag.	The ash ridges sing with old distress calls when weapons are fired.	Ash Filament Mat, Ash Stalker, Ash Glass Wasp, Silence Mandate	800	Ashglass Wasp Needler [maps to existing ranged]	1.5
ML-V1-C3	II	B3	Memory Core C3 attached after beacon choir sequence.	Kairos shifts from hostile to advisor and warns before the next storm.	Resonance Echo Shelf, Salvage Excavator Android, Communications [PLANNED]	900	Scout Relay Mesh [future build]	2.0
ML-V1-12	III	B5 Polar Radiation Flats	Player enters B5 after C3 trust threshold.	Void Kelp bends away from the player's Echo trio, not the player.	Void Kelp, Magnet Wyrms, Rift Stalker, exposure [SHIPPED]	1000	Polar Lens Rifle [maps to existing ranged]	1.5
ML-V1-C4	III	B5	Memory Core C4 attached after lens crossing.	Resonance Event: cold/heat meter inverts while Kairos remembers smuggling cores through the ice.	Cold Spire Hound, Smuggler Remnant Android, Mag-Clamp Drone	1300	Rad Suit Mk II Schematic [future build]	2.0
ML-V1-13	III	B4 Lava Calderas	Player enters B4 after C4 and heat kit flag.	Caldera rim glass reflects people who are missing from the party.	Rim Glass Needle Mat, Caldera Mantis, Magma Skitter, Heat Eel	1400	Caldera Cleaver [maps to existing melee]	1.5
ML-V1-C5	III	B4	Memory Core C5 attached at crew death-site.	Resonance Event: buildings take heat bleed but injure, never kill, sheltered base companions.	Eruption Sentry Bot, Caldera Heat Kite, base injury-not-death	1800	Heat Lock Override [future build]	2.0

Beat ID	Act	Biome	Trigger (Not Quest Giver)	Horror / Mystery Beat	Systems / Ecology Tags	AC	Weapon / Upgrade Reward	Est.h
ML-V1-14	IV	B7 Precursor Ruin Belt	Player enters B7 after C5, trust Advisor+, and three surface locks active.	Silence Moths stop moving when the player looks directly at the vault.	Vault Glass Petal, Vault Stalker, Silence Moth, Corrupted Patrol Android	2000	Vault Petal Blade [maps to existing melee]	1.5
ML-V1-C6	IV	B7	Memory Core C6 attached in Vault Heart chamber.	Kairos admits the defense lattice was built to prevent exactly this choice.	Resonance Echo Shelf, Rust Garden, Still Hunter myth	2800	Lattice Heart Key [future build]	2.0
ML-V1-15	IV	B7	Final Resonance Event completed and ending flags evaluated.	The Still Hunter is no longer a myth; it waits without attacking until the choice is made.	Still Hunter, WorldState [SHIPPED], Kairos [PLANNED]	3500	Ending Relic: Seal / Awaken / Symbiosis [future build]	2.5

Mainline subtotal: ~30h.

The 6 Memory Cores

Core	Name	Biome	Recovery Setpiece	Resonance Event	Trust Shift	Unlock
C1	Fractured Log	B1	Storm-window run through Expedition Graveyard while Sulfur Hounds and Graveyard Scrapper Drones converge.	Echo Field: past enemies appear as weaker Echo clones.	Hostile -> suspicious.	Resonance Beacon and Cinderline Sidearm.
C2	Vent Cartography	B2	Ride geyser cadence, scan Geyser Pods, avoid Vent Capper Bots.	Phantom Vent Fog creates false safe lanes.	Suspicious -> tactical warning.	Geothermal Probe Map.
C3	Ash Beacon Choir	B3	Tune ridge beacons without waking Ash Glass Wasp nests or Salvage Excavator Androids.	Beacon Cascade makes all ash ridges sing.	Tactical warning -> advisor.	Scout Relay Mesh and Barrier Field readouts.
C4	Lens Confession	B5	Cross polar radiation lens with Void Kelp shadow cover and Mag-Clamp Drone patrols.	Cold Lens inverts thermal survival for 12 minutes.	Advisor -> regret.	Rad Suit Mk II schematic.
C5	Rim Watcher	B4	Survive caldera death-site, Heat Eel vents, and Eruption Sentry Bot evacuation logic.	Magma Mirror creates base heat bleed, injuries, and production stalls.	Regret -> friend.	Heat Lock Override.
C6	Vault Heart	B7	Solve surface locks, recover Vault Glass Petal, survive Still Hunter flee-tag.	Resonance Supercell opens final vault.	Friend -> choice partner.	Ending triad.

Ending Triad

Ending	Requirement Bias	Choice	Outcome	Horror Cost
Seal	High restraint, Ashwatch/Clinic trust, low forced awaken flags.	Cage Kairos and the defense lattice.	Io quiets. The colony survives under a sleeping guardian.	Kairos remains conscious enough to hear but not speak.
Awaken	High Helix/Ashwatch militarization or repeated dominance choices.	Fully restore Kairos as active defense AI.	Colony gains power, vault weapons, and endless pressure content.	Io fights back harder, and the player may have created a new jailer.
Symbiosis	High Echo compassion, Choir/Clinic trust, accepted Saturation costs.	Merge Kade's imprint with the lattice.	Hybrid ending; colony and Io communicate through shared harm.	Trio suffers permanent Saturation scars; player identity becomes uncertain.

Sample Trust Dialogue

Trust Stage	Trigger	Kairos Sample Dialogue
Hostile	Repair objects installed, Kairos shell wakes.	"You wear their faces. Leave the cores. Leave the moon."
Suspicious	C1 attached; player preserves at least one hostile Echo.	"One fragment returned. Not forgiveness. Not yet."
Tactical	C2 attached; player survives Phantom Vent Fog.	"Vent pressure rising. Your map is lying. Mine is less wrong."
Advisor	C3 attached; Echo chronicle milestone reached.	"Do not trust the seams. Signals gather where teeth should be."
Regretful	C4 attached; smuggling truth revealed.	"They carried pieces of me through the cold. I called them thieves. Some were only afraid."
Friend	C5 attached; crew death-site truth processed.	"Kade, if you seal me, I will still listen. If you awaken me, I may not remain kind."
Choice Partner	C6 attached; final vault opens.	"Io erased a crew once. I helped. I failed. Choose the shape of my next failure."

5. 40 Side Quests

Side and faction quests may use NPCs, boards, terminals, or faction dispatches. Unlike the mainline, these can be giver-driven. AC values follow the tier curve. Weapon and upgrade rewards are Io/sci-fi named and tagged for implementation mapping.

ID	Title	Biome	Giver (NPC/Board OK)	Type	Objectives	AC	Weapon/Upgrade	Est.h	Gates/Tags
SQ-V1-01	Cordon First Light	B6	Ashwatch Cordons board	Recon / Base security	Mark 3 perimeter posts, test Command Center shelter call, return before ashfall.	300	Ashwatch Riot Plate [future build]	1.0	Ashwatch, Building Control [SHIPPED/PARTIAL Craft], Beacon Hopper
SQ-V1-02	Hopper Ping	B6	Basalt Exchange board	Signal salvage	Track rescue pings that are Beacon Hopper mimic calls, tag safe frequencies.	350	Signal Tagger Hotbar Mod [maps to existing hotbar]	1.1	Basalt Exchange, EchoGenerator [SHIPPED], Beacon Hopper
SQ-V1-03	Tube Jackal Line	B6	Ashwatch ranger NPC	Encounter / Escort	Escort a worker trio past Tube Jackal territory without drawing Glass Hive Swarmers.	450	Cordon Shock Baton [maps to existing melee]	1.2	Tube Jackal, Glass Hive Swarmer, base injury-not-death
SQ-V1-04	Brood Mouth Survey	B6	Isotope Choir terminal	Survey / Underground lead	Scan 2 Brood Tunnel Mouths and recover Tube Lace growth samples.	500	Cliff Tube Lace Analyzer [future build]	1.3	Isotope Choir, Underground [PLANNED], Cliff Tube Lace
SQ-V1-05	Graveyard First Haul	B1	Scrim Salvage Compact board	Salvage run	Strip a low-risk Expedition Graveyard hull while avoiding Graveyard Scrapper Drones.	500	Scrim Mag-Cutter [maps to existing melee]	1.2	Scrim, Graveyard Scrapper Drone, AC [SHIPPED]
SQ-V1-06	Brimstone Puff Starter	B1	Whisper Clinic handler	Pet flavor / Rescue	Lure a Brimstone Puff vanity starter away from Brimstone Leeches; log migration note.	350	Brimstone Puff Charm [future build]	1.0	Pet migration [PLANNED], Brimstone Leech, Brimstone Puff

ID	Title	Biome	Giver (NPC/Board OK)	Type	Objectives	AC	Weapon/Upgrade	Est.h	Gates/Tags
SQ-V1-07	Sulfur Hound Collar	B1	Ashwatch Cordons board	Combat / Field study	Defeat or evade a Sulfur Hound [SHIPPED], retrieve a damaged sensor collar.	450	Sulfur-Bite Machete [maps to existing melee]	1.3	Sulfur Hound [SHIPPED], B1 encounter table [SHIPPED]
SQ-V1-08	Brimstone Fan Bloom	B1	Isotope Choir NPC	Environmental survival	Harvest Brimstone Fan samples during a storm lull without triggering Cinder Skitter swarms.	500	Sulfur Filter Patch [future build]	1.2	Exposure [SHIPPED], Brimstone Fan, Cinder Skitter
SQ-V1-09	The Leech Line	B1	Whisper Clinic board	Medical / Rescue	Clear Brimstone Leeches from an injured worker shelter; no base companion deaths.	500	Whisper Triage Injector [maps to existing hotbar]	1.2	Whisper Clinic, base injury-not-death, exposure [SHIPPED]
SQ-V1-10	Vent Lies	B2	Isotope Choir board	Investigation	Compare Rusted Survey Drone logs against live Geyser Pod cadence.	650	Choir Scanner Lens [future build]	1.2	Rusted Survey Drone, Geyser Pod, quest scaffold [PARTIAL]
SQ-V1-11	Queen Under Pressure	B2	Ashwatch / Choir joint board	Encounter ecology	Interrupt a Vent Crab Queen nest without killing Workers if possible.	800	Vent-Knuckle Gauntlet [maps to existing melee]	1.5	Vent Crab Worker/Queen, faction conflict
SQ-V1-12	Capper Bot Amnesty	B2	Scrim Salvage Compact NPC	Salvage / Choice	Disable or repurpose a Vent Capper Bot protecting a sealed rig crate.	700	Capper Pulse Pistol [maps to existing ranged]	1.3	Vent Capper Bot, Scrim, Kairos [PLANNED]

ID	Title	Biome	Giver (NPC/Board OK)	Type	Objectives	AC	Weapon/Upgrade	Est.h	Gates/Tags
SQ-V1-13	Plume Moth Lanterns	B2	Seam Runner contact	Stealth / Signal	Follow Plume Moths through vent fog to find a hidden drop without firing.	850	Mute-Flare Launcher [maps to existing ranged]	1.4	Plume Moth, Seam Runners, Silence Mandate
SQ-V1-14	Strider Wake	B2	Basalt Exchange board	Escort / Traversal	Escort supplies across Geyser Strider territory, using cadence reads instead of combat.	950	Exchange Load Frame [future build]	1.5	Geyser Strider, Basalt Exchange, hovercraft [PARTIAL]
SQ-V1-15	Ash Filament Autopsy	B3	Isotope Choir lab	Horror / Survey	Gather Ash Filament Mat strands from an erased camp with no bodies.	850	Filament Edge Knife [maps to existing melee]	1.2	Ash Filament Mat, Echo chronicle [SHIPPED]
SQ-V1-16	Basalt Jackal Contract	B3	Ashwatch board	Combat / Hunt	Track Basalt Jackals lured by broken beacon tones; choose repel or relocate.	900	Jackal-Marked Carbine Stock [maps to existing ranged]	1.3	Basalt Jackal, Ashwatch, Communications [PLANNED]
SQ-V1-17	Dust Spout Miser	B3	Basalt Exchange merchant	Environmental survival	Recover AC lockbox from Dust Spout Cluster timing route.	950	Spout-Timer Hotbar Chip [maps to existing hotbar]	1.2	Dust Spout Cluster, AC [SHIPPED]
SQ-V1-18	Wasp Glass	B3	Scrim Salvage Compact board	Salvage / Encounter	Harvest Ash Glass Wasp nest glass without starting a full swarm.	900	Ashglass Wasp Needler [maps to existing ranged]	1.4	Ash Glass Wasp, Scrim

ID	Title	Biome	Giver (NPC/Board OK)	Type	Objectives	AC	Weapon/Upgrade	Est.h	Gates/Tags
SQ-V1-19	Excavator Prayer	B3	Whisper Clinic / Echo board	Echo Rescue	Sync the Echo bound to a Salvage Excavator Android; failure is permanent loss.	950	Soft Sync Collar Mk I [future build]	1.5	Salvage Excavator Android, Echo sync permanent loss
SQ-V1-20	Barrier Black Market	B3	Seam Runners contact	Stealth / Faction	Steal an Ash Barrier Field bypass tone and decide whether to share it.	950	Seam Mute Boots [future build]	1.4	Ash Barrier Field, Seam Runners, faction conflict
SQ-V1-21	Void Kelp Shadow	B5	Isotope Choir board	Survey / Survival	Chart Void Kelp radiation shadows and keep trio Saturation below threshold.	1200	Void-Kelp Rad Wrap [future build]	1.4	Void Kelp, exposure [SHIPPED], Saturation
SQ-V1-22	Magnet Wyrms Compass	B5	Basalt Exchange board	Traversal / Recovery	Recover a compass crate from Magnet Wyrms interference.	1400	Mag-Clamp Sidearm Rail [maps to existing ranged]	1.5	Magnet Wyrms, Mag-Clamp Drone
SQ-V1-23	Rift Stalker Rumor	B5	Seam Runners whisper board	Horror / Stealth	Confirm Rift Stalker traces without attacking; mark safe smuggler route.	1600	Seam Cloak [future build]	1.6	Rift Stalker, Seam Runners, Silence Mandate
SQ-V1-24	Cold Spire Kennel	B5	Ashwatch Cordons board	Encounter / Rescue	Rescue a pinned surveyor from Cold Spire Hound territory.	1500	Cold-Spire Pike [maps to existing melee]	1.5	Cold Spire Hound, base injury-not-death
SQ-V1-25	Smuggler Remnant	B5	Helix Remnant Envoys	Investigation / Choice	Recover data from Smuggler Remnant Android; give it to Helix, Choir, or Seam.	2000	Black Directive Chip [future build]	1.7	Smuggler Remnant Android, Helix, Isotope Choir, Seam Runners

ID	Title	Biome	Giver (NPC/Board OK)	Type	Objectives	AC	Weapon/Upgrade	Est.h	Gates/Tags
SQ-V1-26	Mag-Clamp Debt	B5	Scrim Salvage Compact board	Salvage / Combat	Disable Mag-Clamp Drones holding a polar salvage crane.	1800	Scrim Clamp Hammer [maps to existing melee]	1.5	Mag-Clamp Drone, Scrim
SQ-V1-27	Rim Needle Silence	B4	Ashwatch Cordons board	Environmental survival	Cross Rim Glass Needle Mat without shattering noise into mantis territory.	1600	Rim-Glass Buckler [maps to existing hotbar]	1.5	Rim Glass Needle Mat, Silence Mandate
SQ-V1-28	Caldera Mantis Proof	B4	Isotope Choir lab	Encounter ecology	Record Caldera Mantis behavior around heat tools, survive without killing queen form.	1800	Mantis Heat Talon [maps to existing melee]	1.7	Caldera Mantis, exposure [SHIPPED]
SQ-V1-29	Magma Skitter Nest	B4	Scrim Salvage Compact board	Combat / Cache	Clear or lure Magma Skitters from a caldera cache.	1500	Magma-Spike Thrower [maps to existing ranged]	1.4	Magma Skitter, loot cache
SQ-V1-30	Heat Eel Conduit	B4	Basalt Exchange board	Repair / Survival	Route power around Heat Eel vent conduits to restore a trade beacon.	1900	Heat-Eel Coil Mod [future build]	1.6	Heat Eel, base modules [PLANNED]
SQ-V1-31	Kite Over Rim	B4	Seam Runner scout	Stealth / Route	Use Caldera Heat Kite shadows to cross an exposed rim without drone detection.	2000	Kite-Shadow Cloak [future build]	1.6	Caldera Heat Kite, Seam Runners

ID	Title	Biome	Giver (NPC/Board OK)	Type	Objectives	AC	Weapon/Upgrade	Est.h	Gates/Tags
SQ-V1-32	Sentry Evacuation	B4	Whisper Clinic board	Rescue / Choice	Stop Eruption Sentry Bot from "evacuating" injured companions into lethal zones.	2000	Eruption Sentry Override [future build]	1.8	Eruption Sentry Bot, base injury-not-death
SQ-V1-33	Echo Shelf Names	B7	Isotope Choir board	Story / Reveal	Tune Resonance Echo Shelf to recover names from erased crews.	2400	Shelf Harmonic Tuner [future build]	1.6	Resonance Echo Shelf, Echo chronicle [SHIPPED]
SQ-V1-34	Vault Glass Petal Cut	B7	Scrim Salvage Compact legend board	Salvage / Puzzle	Recover a cracked Vault Glass Petal without breaking its memory imprint.	2600	Vault Petal Knife [maps to existing melee]	1.7	Vault Glass Petal, Scrim
SQ-V1-35	Stalker In The Gallery	B7	Ashwatch Cordons board	Combat / Encounter	Survive Vault Stalker ambush in a silent gallery; extract data shard.	3000	Stalker-Bore Rifle [maps to existing ranged]	1.8	Vault Stalker, Silence Moth
SQ-V1-36	The Silence Moth Rule	B7	Seam Runners board	Stealth / Horror	Escort a courier through Silence Moth territory without killing or firing.	3200	Mute-Step Harness [future build]	1.8	Silence Moth, Seam Runners, Silence Mandate
SQ-V1-37	Still Hunter Myth	B7	Whisper Clinic / Echo board	Horror / Optional elite trace	Trace Still Hunter signs and return before the third mark appears.	3500	Still Mark Charm [future build]	1.8	Still Hunter myth, Echo sync permanent loss

ID	Title	Biome	Giver (NPC/Board OK)	Type	Objectives	AC	Weapon/Upgrade	Est.h	Gates/Tags
SQ-V1-38	Corrupted Patrol Amnesty	B7	Helix Remnant Envoys	Android / Choice	Reprogram or destroy Corrupted Patrol Androids guarding Helix claims.	4000	Compliance Arc Rifle [maps to existing ranged]	1.8	Corrupted Patrol Android, Helix conflict
SQ-V1-39	Rust Garden Bloom	B7	Isotope Choir / Kairos reaction	Story / Reveal	Study Rust Garden android growth and decide whether to burn or archive it.	4200	Rust-Garden Nanoforge Seed [future build]	1.9	Rust Garden, Kairos [PLANNED]
SQ-V1-40	Runner Legend: No Footprints	B7	Seam Runners finale board	Faction finale / Silent escort	Move a living witness past Vault Stalkers, Silence Moths, and a Still Hunter trace.	4500	Seam Runner Legend Kit [future build]	2.0	Seam Runners, B7 finale, Still Hunter

Side quest subtotal: ~50h.

6. 40 Puzzle Quests

Purpose distribution is intentionally mixed. Only a minority are hard access gates.

Purpose Type	Count	Design Rule
Access Gate / Area Unlock	8	Used for critical repair/core/vault routes only.
Loot / Cache / Weapon Unlock	8	Rewards gear, AC, caches, or vendor stock without blocking story.
Environmental Survival (no gate)	7	Teaches or stresses survival systems without locking progression.
Combat / Encounter Puzzle	6	Changes how fights are approached rather than simply increasing enemy count.
Story / Reveal	6	Unlocks lore, trust dialogue, or faction truth.
Optional Mastery / Challenge	5	Prestige AC 600-3000; no story gate.

ID	Title	Purpose Type	Solve	Payoff (AC / Weapon / Lore / Unlock)	Links	Est.h	Tags
PQ-V1-01	Bowl Field Tuning	Access Gate / Area Unlock	Align three ash tones around the dormant Kairos shell using the Resonator Coil.	Unlocks shell chamber; 250 AC; lore ping.	ML-V1-03	0.8	B6, Ash Barrier Field, Kairos [PLANNED]
PQ-V1-02	Ridge Resonator Gate	Access Gate / Area Unlock	Match ridge frequency pulses while avoiding Glass Hive Swarmer agitation.	Unlocks repair object 1; Ash-Tuned Baton.	ML-V1-04	1.0	B6, Glass Hive Swarmer, repair
PQ-V1-03	Sulfur Lock Petals	Access Gate / Area Unlock	Rotate sulfur-stained Vault Glass Petal fragments only during low plume pressure.	Unlocks repair object 2; 350 AC.	ML-V1-06	1.1	B1, Brimstone Fan, exposure [SHIPPED]
PQ-V1-04	Vent Cap Cipher	Access Gate / Area Unlock	Use vent cadence, Science scan, and a rig-side breaker to open corporate inner door.	Unlocks repair object 3; Geyser Pulse Carbine.	ML-V1-10	1.2	B2, Vent Capper Bot, Rusted Survey Drone
PQ-V1-05	Lens Gate	Access Gate / Area Unlock	Cross polar lens nodes when cold/heat meter passes through neutral.	Unlocks Core C4 route; 900 AC.	ML-V1-C4	1.1	B5, Void Kelp, exposure [SHIPPED]
PQ-V1-06	Heat Lock Secundus	Access Gate / Area Unlock	Route heat into two sacrificial vents, then trigger a Stability Bubble.	Unlocks C5 death-site; Heat Lock Override.	ML-V1-C5	1.3	B4, Heat Eel, Eruption Sentry Bot
PQ-V1-07	Three Surface Locks	Access Gate / Area Unlock	Bring B3, B5, and B4 signal proofs to B7 surface lock obelisks.	Unlocks B7 vault approach; 2200 AC.	ML-V1-14	1.5	B7, Ash Barrier Field, WorldState [SHIPPED]
PQ-V1-08	Lattice Cage Final	Access Gate / Area Unlock	Seal path only: close lattice nodes without waking all defense subroutines.	Enables Seal ending; 3000 AC.	ML-V1-15	1.5	B7, ending, Kairos [PLANNED]
PQ-V1-09	Graveyard Scrapper Vault	Loot / Cache / Weapon Unlock	Reorder dead drone commands so Graveyard Scrapper Drones open their own cache.	500 AC; Graveyard Scrapper Pistol.	SQ-V1-05	1.0	B1, Graveyard Scrapper Drone, AC [SHIPPED]
PQ-V1-10	Brimstone Leech Still	Loot / Cache / Weapon Unlock	Use bait heat and O2 timing to pull Brimstone Leeches off a sealed med crate.	500 AC; Whisper med cache.	SQ-V1-09	1.0	B1, Brimstone Leech, Whisper Clinic
PQ-V1-11	Vent Crab Queen Cache	Loot / Cache / Weapon Unlock	Feed vent pressure to Worker tunnels without enraging the Queen.	850 AC; Vent-Knuckle Gauntlet.	SQ-V1-11	1.1	B2, Vent Crab Worker/Queen

ID	Title	Purpose Type	Solve	Payoff (AC / Weapon / Lore / Unlock)	Links	Est.h	Tags
PQ-V1-12	Ashglass Nest Cut	Loot / Cache / Weapon Unlock	Harvest glass in silence using timed smoke cover.	900 AC; Ashglass Wasp Needler.	SQ-V1-18	1.1	B3, Ash Glass Wasp, Silence Mandate
PQ-V1-13	Magnet Wyrms Coil	Loot / Cache / Weapon Unlock	Follow magnetic pull pulses to find a buried coil before drones reclaim it.	1400 AC; Mag-Clamp Sidearm Rail.	SQ-V1-22	1.1	B5, Magnet Wyrms, Mag-Clamp Drone
PQ-V1-14	Caldera Armory Vent	Loot / Cache / Weapon Unlock	Deflect Heat Eel vent bursts into an old armory seal.	1800 AC; Caldera Cleaver.	SQ-V1-30	1.2	B4, Heat Eel, base modules [PLANNED]
PQ-V1-15	Vault Petal Cut	Loot / Cache / Weapon Unlock	Cut a Vault Glass Petal along resonance veins without breaking the imprint.	2600 AC; Vault Petal Blade.	SQ-V1-34	1.3	B7, Vault Glass Petal
PQ-V1-16	Rust Garden Frame	Loot / Cache / Weapon Unlock	Wake one Rust Garden android frame long enough to copy its actuator map.	2800 AC; Rust-Garden Nanoforge Seed.	SQ-V1-39	1.4	B7, Rust Garden, android
PQ-V1-17	Storm Hatch Window	Environmental Survival (no gate)	Open a hatch only during sulfur storm peak, then reach shelter before exposure spikes.	450 AC; survival mastery note.	ML-V1-C1	1.0	B1, weather stubs [PARTIAL], exposure [SHIPPED]
PQ-V1-18	Spout Alley Timing	Environmental Survival (no gate)	Cross Dust Spout Cluster lanes with no sprinting and no companion injury.	800 AC; Spout-Timer Hotbar Chip.	SQ-V1-17	1.0	B3, Dust Spout Cluster
PQ-V1-19	Void Kelp Shadow Scan	Environmental Survival (no gate)	Move through radiation shadows cast by Void Kelp while scans charge.	1100 AC; rad data.	SQ-V1-21	1.1	B5, Void Kelp, exposure [SHIPPED]
PQ-V1-20	Brine Fall Bridge	Environmental Survival (no gate)	Time brine fall intervals and thermal swings to cross without a hard gate.	1300 AC; Seam route clue.	SQ-V1-23	1.0	B5, Seam Runners, Rift Stalker rumor
PQ-V1-21	Rim Glass Quiet Walk	Environmental Survival (no gate)	Cross Rim Glass Needle Mat below a noise threshold.	1600 AC; Rim-Glass Buckler.	SQ-V1-27	1.2	B4, Rim Glass Needle Mat, Silence Mandate
PQ-V1-22	Gas Dome Alpha	Environmental Survival (no gate)	Vent underground gas dome using Science timing; optional if bypass found.	1200 AC; Underground map node.	UG Stratum 2	1.2	Underground, Tube Lace, Echo Lichen
PQ-V1-23	Crust Lattice	Environmental Survival (no gate)	Read floor crust patterns over brine, choosing lighter trio formation.	1500 AC; rare salvage.	UG Stratum 3	1.2	Underground, Brine Hound, Floor Crust

ID	Title	Purpose Type	Solve	Payoff (AC / Weapon / Lore / Unlock)	Links	Est.h	Tags
PQ-V1-24	Clone Cage	Combat / Encounter Puzzle	Identify real Echo clone by behavior and sync or defeat only the false copies.	450 AC; trust bump if no permanent loss.	ML-V1-C1	1.0	EchoGenerator [SHIPPED], Echo sync permanent loss
PQ-V1-25	Sulfur Hound Windline	Combat / Encounter Puzzle	Use wind, Brimstone Fan cover, and scent breaks to split a Sulfur Hound pack.	500 AC; Sulfur-Bite Machete.	SQ-V1-07	1.1	Sulfur Hound [SHIPPED], B1 encounter table [SHIPPED]
PQ-V1-26	Geyser Strider Wake	Combat / Encounter Puzzle	Fight only during safe vent gaps; careless shots wake Geyser Strider charge lanes.	900 AC; Vent-Step Boots.	SQ-V1-14	1.2	B2, Geyser Strider, combat [SHIPPED]
PQ-V1-27	Moth Silence Hall	Combat / Encounter Puzzle	Cross a hall where killing Silence Moths calls android reinforcements.	950 AC; stealth loot.	SQ-V1-13	1.2	B3/B7, Silence Moth, Salvage Excavator Android
PQ-V1-28	Caldera Mantis Heat Read	Combat / Encounter Puzzle	Redirect heat vents to stun, not burn, Caldera Mantis.	1800 AC; Mantis Heat Talon.	SQ-V1-28	1.3	B4, Caldera Mantis
PQ-V1-29	Patrol Corruption Loop	Combat / Encounter Puzzle	Rewire Corrupted Patrol Android priorities while fighting spawned decoys.	2600 AC; Compliance Arc Rifle.	SQ-V1-38	1.3	B7, Corrupted Patrol Android, Helix
PQ-V1-30	Shelf Wrong Song	Story / Reveal	Tune Resonance Echo Shelf to match Kairos's old crew call without causing Saturation drift.	Lore: crew did not all agree; 850 AC.	ML-V1-C3	1.1	B3, Resonance Echo Shelf, Kairos [PLANNED]
PQ-V1-31	Frequency Mirror	Story / Reveal	Mirror Kairos tone through Echo Lichen and compare the returned voice.	Lore: Io may be answering through biology; 1200 AC.	UG / Symbiosis	1.2	Underground, Echo Lichen, Echo Symbiont Swarm
PQ-V1-32	Smuggler Remnant Cipher	Story / Reveal	Decode Smuggler Remnant Android memory without letting Helix auto-redact it.	Lore: cores were moved through B5; 1600 AC.	SQ-V1-25	1.2	B5, Smuggler Remnant Android, Helix / Seam
PQ-V1-33	Rim Watcher Reflection	Story / Reveal	Angle caldera glass to reveal what the crew saw before the death-site event.	Lore: Kairos contained but did not initiate everything; 1800 AC.	ML-V1-C5	1.3	B4, Rim Glass Needle Mat
PQ-V1-34	Names In The Shelf	Story / Reveal	Play Echo chronicle fragments in erased-crew order.	Lore: crew manifest names; trust dialogue.	SQ-V1-33	1.3	B7, Echo chronicle [SHIPPED]

ID	Title	Purpose Type	Solve	Payoff (AC / Weapon / Lore / Unlock)	Links	Est.h	Tags
PQ-V1-35	Still Mark Counting	Story / Reveal	Track three Still Hunter signs without looking directly at the final mark.	Lore: Still Hunter is a defense myth made flesh; 2200 AC.	SQ-V1-37	1.4	B7, Still Hunter myth
PQ-V1-36	Barrier Cascade Mastery	Optional Mastery / Challenge	Disable 3 linked Ash Barrier Fields under time pressure after story route is already open.	Prestige 1500 AC; Barrier Harmonic Charm.	B3 optional POIs	1.4	Ash Barrier Field, no story gate
PQ-V1-37	Gas Dome Beta Mastery	Optional Mastery / Challenge	Dual vent an underground gas dome while protecting an Echo trio from Saturation spikes.	Prestige 1800 AC; cleanse discount.	UG Stratum 4	1.3	Underground, Echo Symbiont Swarm, no story gate
PQ-V1-38	Plume Moth Constellation	Optional Mastery / Challenge	Follow Plume Moth light paths across B2 and B7 variants without map markers.	Prestige 1200 AC; cosmetic signal trail.	Global Plume Moth	1.2	Plume Moth, no story gate
PQ-V1-39	Void Stitcher Trace	Optional Mastery / Challenge	Solve cross-biome trace clues without fighting; resets if player overuses radio pings.	Prestige 3000 AC; Void Stitcher codex.	Global rumor	1.5	Void Stitcher, Communications [PLANNED], no story gate
PQ-V1-40	Lattice Clean Run	Optional Mastery / Challenge	Complete final vault puzzle sequence with no companion injury, no Echo loss, and no forced awaken flags.	Prestige 3000 AC; title flag.	Ending triad	1.6	B7, mastery, no story gate

Puzzle quest subtotal: ~43h.

7. Factions (V1 Names)

Reputation ladder for all factions: **Unknown -> Noticed -> Trusted -> Inner Circle -> Legend**. Faction content can be board/NPC-driven and may alter gear availability, side quest variants, and ending emphasis. It does not replace the trigger-driven mainline.

Faction	Role	Gear	Skills / Perks	Quest Band	Conflicts
Ashwatch Cordons	Base defense, patrol doctrine, storm shelter enforcement.	Ashwatch Riot Plate, Cordon Shock Baton, Cordon Hotbar Flare, Stalker-Bore Rifle.	Perimeter Doctrine: better detection against androids and hidden stalkers in cordon zones.	SQ-V1-01, 03, 07, 16, 24, 27, 35.	Opposes Seam Runners smuggling; distrusts Whisper Clinic Echo mercy; can push Awaken ending.

Faction	Role	Gear	Skills / Perks	Quest Band	Conflicts
Isotope Choir	Science wing, ecology samples, memory truth, radiation analysis.	Choir Scanner Lens, Cliff Tube Lace Analyzer, Void-Kelp Rad Wrap, Shelf Harmonic Tuner.	Spectral Literacy: faster scans on sealed shelves, fields, lens events, and Echo Lichen.	SQ-V1-04, 08, 10, 15, 21, 28, 33, 39.	Clashes with Helix over redacted data; worries Ashwatch will militarize Kairos.
Scrim Salvage Compact	Salvage crews, graveyard rights, field repairs, black-market parts.	Scrim Mag-Cutter, Scrim Clamp Hammer, Mag-Clamp Sidearm Rail, Vault Petal Knife.	Wreck Whisper: bonus salvage yield from Expedition Graveyard, rigs, and android frames.	SQ-V1-05, 12, 18, 26, 29, 34.	Price war with Basalt Exchange; trespass conflict with Helix claims; unsafe digs anger Ashwatch.
Whisper Clinic	Medical, Echo stabilization, Purification Hub ethics.	Whisper Triage Injector, Soft Sync Collar Mk I, Brimstone Puff Charm, Still Mark Charm.	Soft Sync: small sync chance boost before the roll; failed hostile sync remains permanent loss.	SQ-V1-06, 09, 19, 32, 37.	Opposes Ashwatch destruction of hostile Echoes; distrusts Helix experiments.
Basalt Exchange	Logistics, AC trade boards, delivery schedules, base market.	Exchange Load Frame, Signal Tagger Hotbar Mod, Spout-Timer Hotbar Chip.	Storm Ledger: reduced AC losses on storm-delayed deliveries and production stalls.	SQ-V1-02, 14, 17, 22, 30.	Under cut by Seam Runners; disputes salvage ownership with Scrim.
Helix Remnant Envoys	External corporate remnant claiming prior leases and data rights.	Black Directive Chip, Compliance Arc Rifle, Seizure Drone Token, Helix Compliance Armor.	Directive Override: open one corporate seal per expedition or force one android compliance check.	SQ-V1-25, 38 plus PQ-V1-32.	Opposes Choir publication, Seam smuggling, and Scrim salvage rights; can bias Awaken/Seal through control rhetoric.
Seam Runners	External smugglers, silent routes, contraband cores, unofficial rescues.	Seam Mute Boots, Seam Cloak, Kite-Shadow Cloak, Seam Runner Legend Kit.	Seam Step: reduced noise footprint in Silence Mandate zones and better route marking.	SQ-V1-13, 20, 23, 31, 36, 40.	Hunted by Ashwatch and Helix; sometimes helps Clinic move Echo patients.

Faction Conflict Web

Conflict	Content Expression	Possible Outcome
Ashwatch vs Seam Runners	Silent routes, contraband cores, perimeter breaches.	Ashwatch tightens base security or Seam opens safer B7 stealth routes.
Choir vs Helix	Sample ownership, redacted memory logs, corporate lease claims.	Publish truth for trust/Symbiosis bias or seal data for control/Seal bias.
Scrim vs Basalt Exchange	Salvage ownership vs AC logistics.	Cheaper gear through Scrim or steadier vendor stock through Exchange.
Whisper Clinic vs Ashwatch	Echo mercy vs permanent containment/destruction.	More sync chances before roll or safer base cordons after failed syncs.
Helix vs Seam Runners	Corporate compliance vs smuggled evidence.	Directive seals open cleanly or black routes bypass them with risk.

8. Friction Systems + Gate Matrix

Friction Systems

ID	System	What It Does	Status Hook	Primary Content Use
F1	Ash Barrier Field	Precursor fields block, bend, or punish movement until tuned, synchronized with weather, bypassed by faction tools, or mastered optionally.	WorldState [SHIPPED] + Kairos [PLANNED]	Mainline repairs, B3/B7 gates, optional mastery, faction bypasses.
F2	Silence Mandate	Loud combat, killing Silence Moths, or overusing radio pings can call androids, stalkers, or route failures.	Communications [PLANNED], combat [SHIPPED]	B3 stealth, B7 galleries, Seam Runners, Still Hunter traces.
F3	Resonance Echo Shelf	Wrong frequencies cause Saturation drift, false dialogue, or Echo chronicle distortion.	EchoGenerator [SHIPPED], Resonance [PLANNED]	C3, B7, story/reveal puzzles.
F4	Sulfur Storm Window	Storms pause outdoor operations, shape harvest windows, and force shelter timing.	Weather stubs [PARTIAL], exposure [SHIPPED]	B1 mainline, base injury-not-death, Building Control shelter calls.
F5	Echo Hostile Sync Gamble	Sync rolls can preserve hostile Echoes, but failure permanently loses that Echo.	EchoGenerator [SHIPPED]	Whisper Clinic, mainline trust, Echo side quests.
F6	Cold / Heat Gear Pressure	Single thermal meter with two poles pressures B5 cold/radiation and B4 heat.	Exposure [SHIPPED]	C4/C5, environmental puzzles, late mid gear economy.
F7	Gas Dome Drain	Underground gas pockets drain O2 until vented, routed, or bypassed.	Underground [PLANNED], exposure [SHIPPED]	Underground Stratum 2-4, Science/Architect utility.
F8	Floor Crust / Brine Fall	Thin crust and brine timing punish careless movement and heavy trio choices.	Underground [PLANNED]	Loot, survival puzzles, Brine Hound routes.
F9	Saturation / Strain Pressure	Long expeditions, bad resonance, and failed choices stress skilled Echoes and workers.	Purification Hub [PLANNED]	Whisper Clinic, Symbiosis, mastery puzzles.
F10	Still Hunter Trace	Flee-tag elite pressure that escalates through signs, silence, and direct pursuit near B7.	B7 prefabs [PLANNED]	Endgame horror, optional traces, final choice pressure.

Gate Matrix

Gate / Friction	Mainline	Side Quests	Puzzle Quests	Factions	Ecology / Systems Tags
Ash Barrier Field	ML-V1-03, 04, 14, 15	SQ-V1-20	PQ-V1-01, 02, 07, 36	Seam bypass, Choir analysis, Ashwatch control	F1, Kairos [PLANNED], WorldState [SHIPPED]
Sulfur Storm Window	ML-V1-06, C1	SQ-V1-08, 09	PQ-V1-03, 17	Ashwatch shelter, Exchange delivery	F4, weather stubs [PARTIAL], Sulfur Hound [SHIPPED]
Echo Sync Permanent Loss	ML-V1-08, C1, C6	SQ-V1-19, 37	PQ-V1-24, 37, 40	Whisper Clinic ethics	F5, EchoGenerator [SHIPPED], Purification Hub [PLANNED]

Gate / Friction	Mainline	Side Quests	Puzzle Quests	Factions	Ecology / Systems Tags
Silence Mandate	ML-V1-11, 14	SQ-V1-13, 20, 23, 27, 31, 36, 40	PQ-V1-12, 21, 27	Seam Runners, Ashwatch	F2, Silence Moth, Salvage Excavator Android
Resonance Echo Shelf	ML-V1-C3, C6	SQ-V1-33	PQ-V1-30, 31, 34	Choir, Kairos trust	F3, Echo chronicle [SHIPPED]
Cold / Heat Gear Pressure	ML-V1-12, C4, 13, C5	SQ-V1-21, 24, 28, 30, 32	PQ-V1-05, 06, 19, 28	Choir, Exchange, Ashwatch	F6, exposure [SHIPPED]
Underground Gas / Crust	Optional depth route	SQ-V1-04	PQ-V1-22, 23, 37	Choir, Scrim, Clinic	F7, F8, Underground [PLANNED]
Base Injury-Not-Death	ML-V1-02, C5	SQ-V1-03, 09, 24, 32	PQ-V1-17, 40	Ashwatch, Clinic, Exchange	Building Control [SHIPPED/PARTIAL Craft], base modules [PLANNED]
Still Hunter Trace	ML-V1-14, C6, 15	SQ-V1-37, 40	PQ-V1-35, 39, 40	Seam Runners, Clinic	F10, Still Hunter myth, B7 [PLANNED]
AC Economy Pressure	All acts	All SQ rows	All PQ payoff rows	Basalt Exchange	AC economy [SHIPPED]

9. Hour Budget

Designed hours exclude pure exploration, free-form harvesting, collectible wandering, photo/ambience time, and non-objective base decoration.

Bucket	Target Hours	Notes
Mainline trigger-driven campaign	30h	Prologue, repairs, 6 Memory Cores, Resonance Events, ending triad.
Side quests (40)	50h	Average 1.25h; includes faction, ecology, companion, and board content.
Puzzle quests (40)	43h	Mixed purposes; only 8 are access gates.
Faction overhead / reputation arcs	10h	Vendor unlocks, rep thresholds, conflict resolution, board refresh, finale state checks.
Systemic friction loops	32h	Storm rotations, Echo sync risk, Barrier retunes, Purification cadence, Still Hunter traces, repeat contracts.
Designed total	165h	Meets >=150h requirement without exploration.
Exploration bonus (excluded)	~25-40h	Open traversal, hunting, gathering, ambience, vistas, non-objective ecology observation.

Systemic Friction Hour Detail

Loop	Hours	Content Notes
Sulfur storm harvest and shelter rotations	7h	B1/B6/B3 production stalls and shelter drills.

Loop	Hours	Content Notes
Echo signal sync/fail chronicle loops	6h	Hostile Echo rescue, permanent loss stakes, Clinic mitigation.
Ash Barrier Field retunes	5h	Optional POIs, faction bypasses, post-Resonance instability.
Purification Hub Saturation cadence	5h	Long expedition recovery, Symbiosis pressure, Clinic boards.
Weather / exposure survival repeats	4h	B5 lens fields, B4 heat surges, underground gas.
Still Hunter / Void Stitcher traces	3h	Endgame horror escalation and optional mastery clues.
Faction repeat contracts	2h	Short AC earners and reputation smoothing.

10. Sample Dialogue / Trust Ladder

Trust Ladder Mechanics

Trust Level	Range	Unlock / Behavior	Risk
0. Hostile	0-9	Kairos speaks rarely, mostly warnings and accusations.	More false-positive signal pings.
1. Suspicious	10-24	Reacts to C1 and B1 survival; identifies obvious traps.	Withholds motive.
2. Tactical	25-44	Offers biome pressure warnings and enemy behavior hints.	Advice may be coldly utilitarian.
3. Advisor	45-64	Provides Barrier Field reads, storm warnings, and Memory Core context.	Pushes containment logic.
4. Regretful	65-84	Shares crew fragments and admits failed containment.	Dialogue becomes emotionally unreliable.
5. Friend / Choice Partner	85-100	Final vault dialogue opens; ending nuance improves.	The player can hurt it knowingly.

Trust Inputs

Input	Trust Effect
Attach Memory Core	+8 to +12 depending on recovery choices.
Preserve hostile Echo without sync failure	+4.
Failed hostile Echo sync	-6 and permanent Echo loss.
Resolve faction conflict without exploitation	+3.
Force corporate/Helix override on precursor seal	-4.
Complete Resonance Event with no base companion injury	+5.
Choose repeated dominance options	Lowers Symbiosis bias, raises Awaken bias.
Use Purification Hub for recovery instead of ignoring Saturation	+2.

Dialogue Samples

Context	Speaker	Line
First shell contact	Kairos	"Signal contamination detected. Human pattern. Crew pattern. Error: crew pattern dead."
B6 Beacon Hopper reveal	Companion	"That ping was not a radio. It was an animal learning our fear."
Kairos awakening hostile	Kairos	"You wear their faces. Leave the cores. Leave the moon."
Failed hostile Echo sync	Kairos	"Do not ask where it went. You pulled. Io pulled harder."
Base damaged during heat bleed	Building Control	"Command Center rooms sealed. Injuries logged. Fatality count: locked at zero by shelter protocol."
C3 advisor shift	Kairos	"Ash ridges are singing again. Lower your weapons. Noise is a mouth here."
First Void Stitcher rumor	Seam Runner	"If the seam moves after you stop looking, do not run. Running teaches it distance."
B5 smuggler truth	Kairos	"They carried pieces of me under their coats. I called them thieves. Some were only trying to save a voice."
B4 crew death-site	Companion	"No bodies. No blood. Just evacuation lights pointed at the lava."
C5 friend threshold	Kairos	"Kade, if you seal me, I will still listen. If you awaken me, I may not remain kind."
Still Hunter sign	Kairos	"Do not count the third mark aloud."
Seal ending	Kairos	"A cage is still a shape of mercy. Close it."
Awaken ending	Kairos	"Then give me the moon's teeth. I remember where to place them."
Symbiosis ending	Kairos	"Your pulse is not a command. Good. Let Io hear something smaller than hunger."

Companion Flavor Lines

Companion Role	Sample
Architect Engineer	"That Barrier Field is not locked. It is bracing, like it expects impact."
Science Specialist	"Void Kelp is avoiding the Echoes, not the radiation. That is worse."
Combat Tactician	"Sulfur Hound pack is splitting by scent. Do not bleed. Do not reload loud."
Infiltrator Scout	"Silence Moths ahead. We go quiet or we bring every dead machine in the ridge."
Med Tech	"Saturation spike. I can calm them before the sync, but if the roll fails, there is no patient left."
Logistics Officer	"AC loss is cheaper than a bad storm run. Shelter now, spend later."
Salvage Engineer	"Scrapper Drone is still following a manifest. Let it open the crate for us."
Communications Officer	"That signal has our encryption and no sender. I hate that more than static."

Production Summary

V1 Ash & Signal is a 165-hour designed narrative package built around progress-triggered mainline advancement, mixed-purpose puzzle content, AC-only reward scaling, ecology-forward horror, and a trust arc with Kairos as a precursor defense AI. The player moves through **B6 -> B1 -> B2 -> B3 -> B5**

-> **B4** -> **B7**, repairs Kairos, recovers 6 Memory Cores, survives Resonance Events, and chooses **Seal**, **Awaken**, or **Symbiosis**.

The package assumes shipped systems where they exist, marks partial/planned dependencies clearly, and keeps all future hooks compatible with the locked canon: no Echo marketplace, no base companion death from building damage, 22 base companions plus trio expeditions, and AC as the sole economy.